

Aryamaan Borgohain

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EDUCATION

Tezpur University

Integrated MSc. in Mathematics
CGPA/SGPA: 7.55/10 (In 1st Semester)

2025 – Present
Tezpur, India

Higher Secondary (AISSCE)

Passed AISSCE with 90.0%, Central Board of Secondary Education
Shrimanta Shankar Academy, Dispur, Guwahati

2025

Matriculation (AISSE)

Passed AISSE with 89.0%, Central Board of Secondary Education
Shrimanta Shankar Academy, Dispur, Guwahati

2023

PROJECTS

OrbitForge: Celestial Mechanics Space-Flight Simulator

- Developed a high-accuracy orbital simulator in C++ and Python to solve Lambert's Boundary Value Problem.
- Implemented Encke's perturbations and higher-order symplectic integration to precisely calculate satellite and rocket launch windows.

Numerical IVP Solvers & ODE Analysis Tools

- Built custom Python/Matplotlib engines to estimate Initial Value Problems (IVPs) and visualize direction fields for autonomous ODEs.
- Authored an independent (unpublished) paper analyzing Forward Euler, Picard's, Taylor series among other approximation methods for IVPs and Autonomous systems.

3D Surface Normal & Gradient Visualization Engine

- Computed partial derivatives and mapped gradient vector fields for explicitly defined 3D surfaces.
- Built a custom, multi-core frame-by-frame animation engine using Python, SciPy, NumPy, and OpenCV.

3D Mobile Skill Development Van Design

- Engineered a comprehensive 3D structural visualization
- Included interior modeling and demo animations, of a modified vehicle chassis for a commercial agency.

SKILLS

Scientific Modules/Tools: Matlab, Spyder IDE, Anaconda, Manim, Matplotlib, SciPy, SymPy, NumPy, Unity 3D

Languages : C++, Python, LaTeX, C#, SQL (MySQL)

Basic Tools : Microsoft 365 Suite (Word, PowerPoint, Excel)